

GitHub repository where all documentation & code can be found. [billbill100 \(github.com\)](https://github.com/billbill100)

ESP32 Shutter Tester pcb V1.1 06/08/2026

As an alternative to using the aliexpress screw-terminal expansion board to mount the ESP32 module, a bespoke pcb has been designed.

All the hard work is done, just got to pcbway from the link below to order the V2 pcb.

[The Shutter Tester - Share Project - PCBWay](#)

Click 'add to basket'. Note the price for 10 boards is the same as for 5. With just a slight postage increase.

The pcb is very similar in design to the aliexpress screw-terminal expansion board, but has the advantage that both the ESP32 module and the tft screen plug directly into the pcb, with all required connections for the tft routed directly on the pcb.

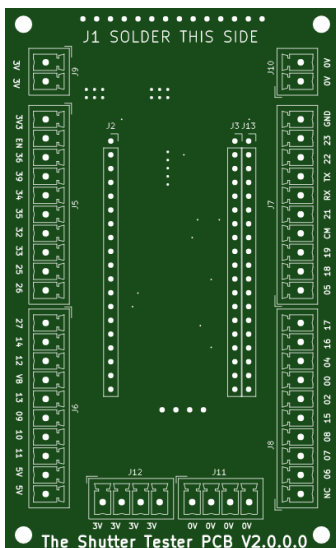
The board (V1 or V2) also has additional 3, 5 and 0v connectors, making wiring simpler.

All of the other connections for sensors etc are broken out onto screw-terminals in the same way as the aliexpress expansion board.

Wiring and using the pcb is identical to that of the aliexpress expansion board, but eliminating the 14 wires for the tft.

The pcb requires header sockets and screw terminals, to be assembled and soldered by the user.

The updated V2 board allows the either the standard NodeMCU32 dev board with pin row spacing of 22.86mm 0.9in or the wider 25.4mm 1.0in board to be used.

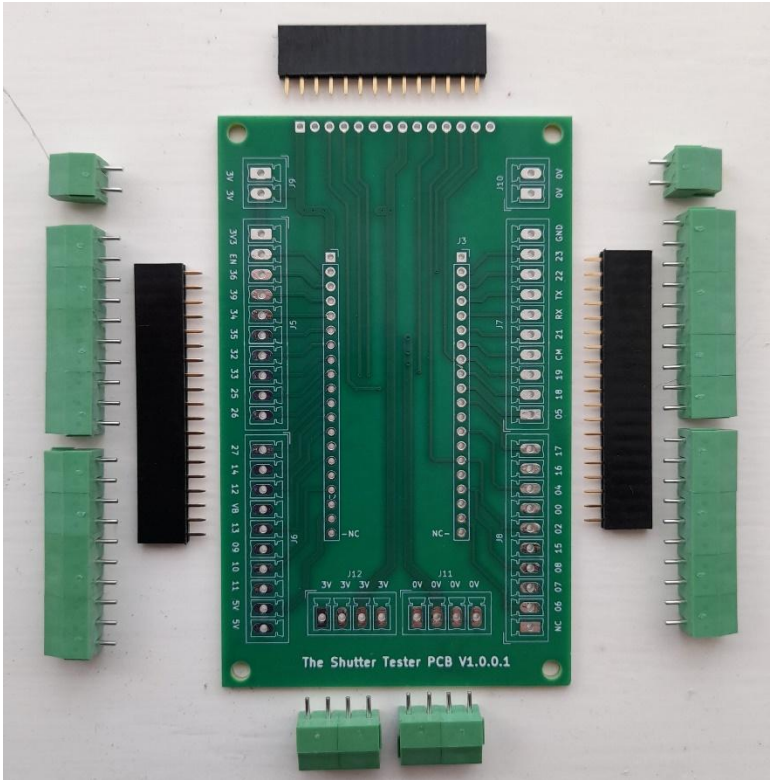


Add the 19-pin header connector in position J3 for the narrower, **OR** J13 for the wider ESP32 dev board.

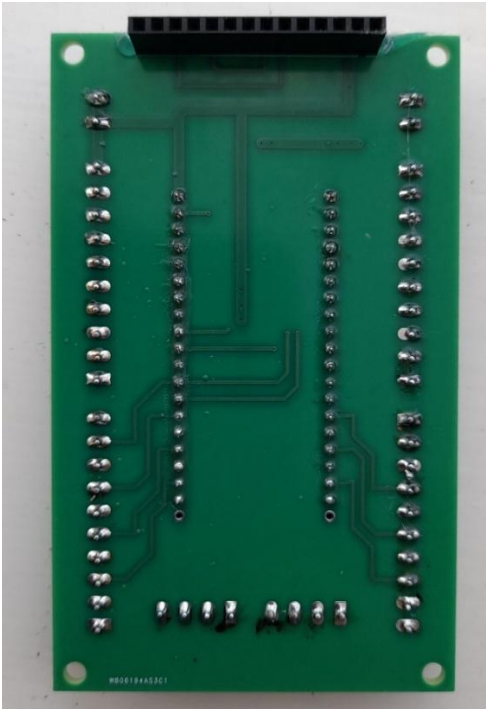
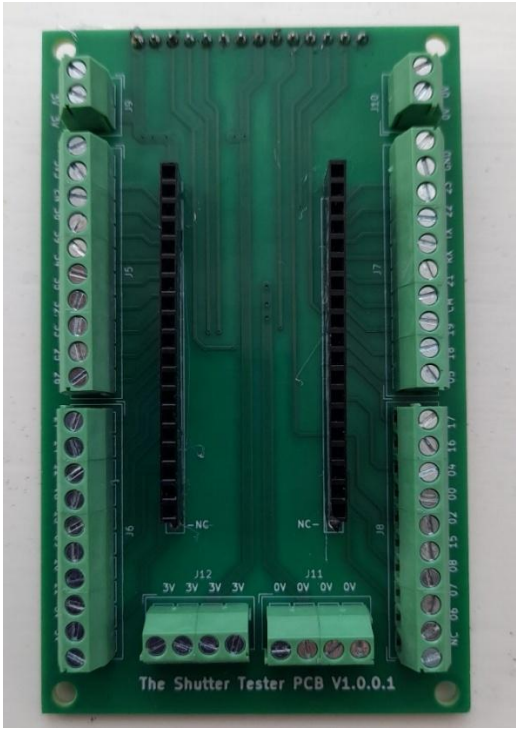
VERY IMPORTANT DISCLAIMER

The V1 board has been ordered and used successfully without error or issue.

The V2 board has NOT yet been tested, whilst there should not be any errors on it (only J13 was added) no responsibility can be taken should an unforeseen error be found with the board.



pcb and additional parts required parts. (V1 board shown)



Completed pcb, showing each side (V1 board shown)

Extra 3v and 0v screw terminals can be seen at the bottom centre of the pcb

Extra 3v and 0v screw terminals top left and right.

Extra 5v screw terminal bottom left.

Important Note:- Ensure the 14-pin tft connector is placed on the rear side of the pcb and the 19-pin ESP32 connectors and screw terminals are placed on the front side of the pcb.

Populate J3 OR J13 with a single 19-pin header connector. DO NOT put a connector in both.

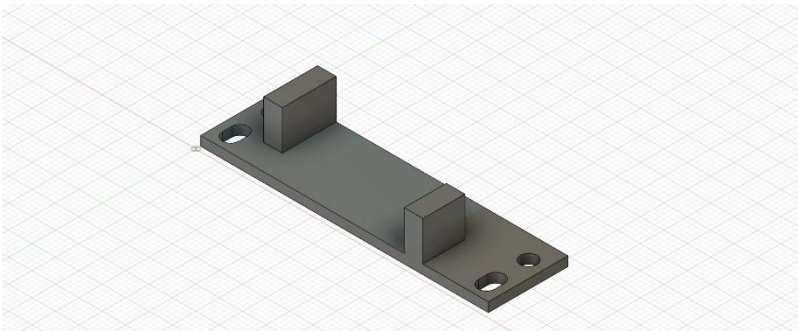
The pre-drilled holes at the top of the pcb align with the holes in the tft, and can be secured using the same tft mounting bolts.

M3 bolts & nuts can be used in the bottom holes, as stand-offs. With the heads hot-glued to the tft pcb. This method is shown in the 'Build a Box' document

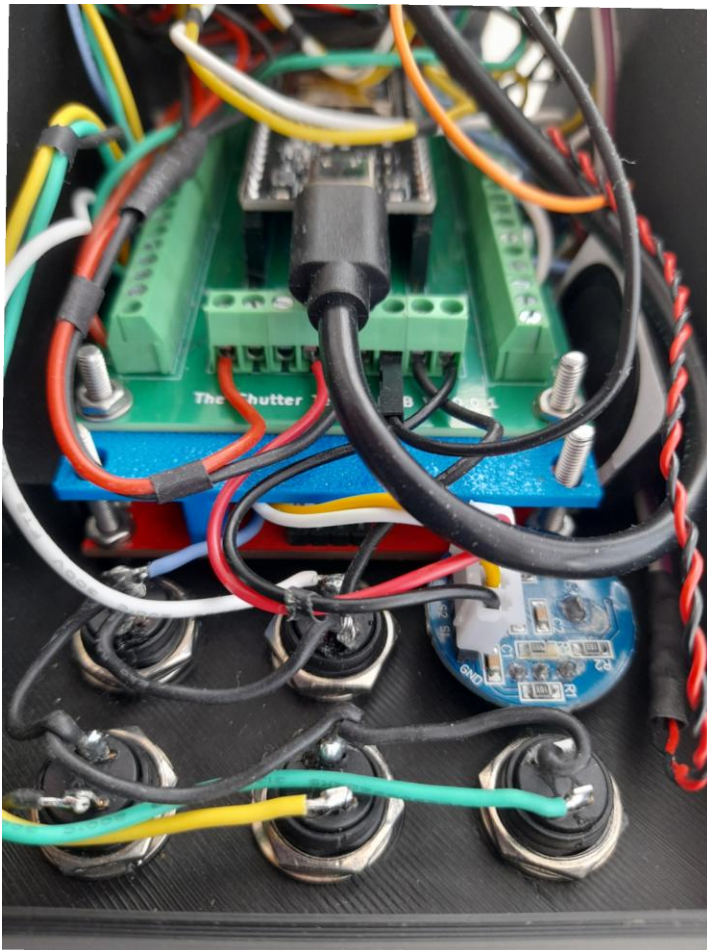
A better solution is to 3d print the tft pcb spacer.

This bolts onto the underside of the pcb, with the rectangular stand-offs facing down, (using the round holes on the 3d printed part) using M3 X12mm bolts or similar.

with the elongated holes fitting onto the tft mounting bolts.



3d printed pcb spacer



Pcb spacer, printed in blue, fitted in position.

Parts required for the pcb

Screw terminals, 3.5mm spacing.

[100PCS 3.5mm 2P 3P Screw Terminal Block Connector KF350](#)

Screw terminals, 3.5mm spacing.

They come in 2 or 3 way and slide together to form larger ways.

4 x 10-way, 2 x 4-way and 2 x 2-way are required.

So one pack of 2-way connectors are required, which is enough for three boards.

(buy 1 pack of 2-way and one pack of 3-way, which will be plenty, if making more than 3)



Header sockets, female.

[10PCS 2.54mm Single Row Female Header 1x2p to 1x40p Pin Connector](#)

2 of 19-way required and

1 of 14-way required.

They come in packs of 10, so one pack of the 19-way will make 5 boards.

(buy 2 packs of 19-way to make 10 boards)

